



KLE'S
Department of Electronics and Communication
Engineering
K. L. E. INSTITUTE OF TECHNOLOGY,
HUBBALLI



A Presentation on
**Design and analysis of rectangular microstrip patch
antenna using antenna magus**
GROUP 36

Presented by

Under the Guidance of
Mrs. Kavita .V.K

Varshinee A Dandavati	2KE23EC180
Sneha salagar	2KE23EC146
Shweta kabanur	2KE23EC142
Ningamma Yaragoppad	2KE23EC075

Literature Survey

S.No	Author & Year	Research Work	Key Findings
1	K. Vidhya et al., 2020	Design and analysis of rectangular microstrip patch antenna using antenna magus	Achieved good return loss, gain and radiation characteristics for wireless applications
2	Shreyashree Verma et al., 2023	Rectangular microstrip patch antenna for 2.4 GHz communication	Improved VSWR, bandwidth and antenna efficiency
3	Sarika Khurana et al., 2017	Simulation of single and stacked rectangular patch antennas	Compared antenna performance and obtained better gain
4	Sattar Othman Hasan et al., 2023	Analysis of different feeding techniques for rectangular patch antenna	Feeding methods improved impedance matching and bandwidth for 5G applications

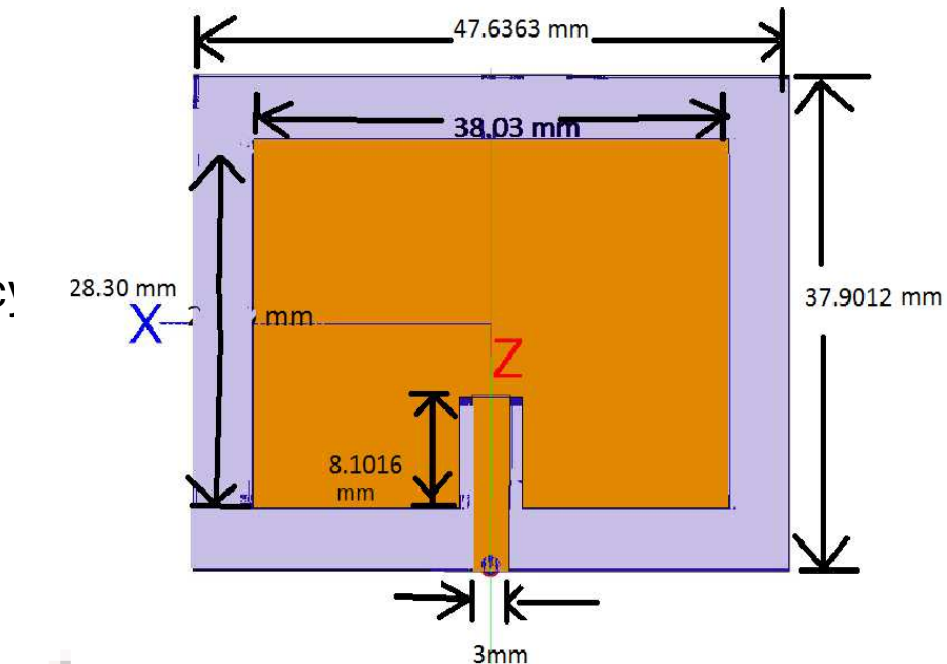
Design of rectangular microstrip patch antenna:

Antenna consists of:

- Radiating patch(copper)
- Dielectric substrate
- Ground plane
- Designed for 2.4GHz frequency
- Substrate thickness:3mm

Substrate materials:

- FR4($\epsilon_r=4.4$)
- RO4003($\epsilon_r=3.4$)
- GML1000($\epsilon_r=3.2$)
- RT/Duroid 5880($\epsilon_r=2.2$)



Design Equations:

$$1. \quad W = \frac{c}{2f_0} * \sqrt{\left(\frac{Z}{\epsilon_r + 1}\right)}$$

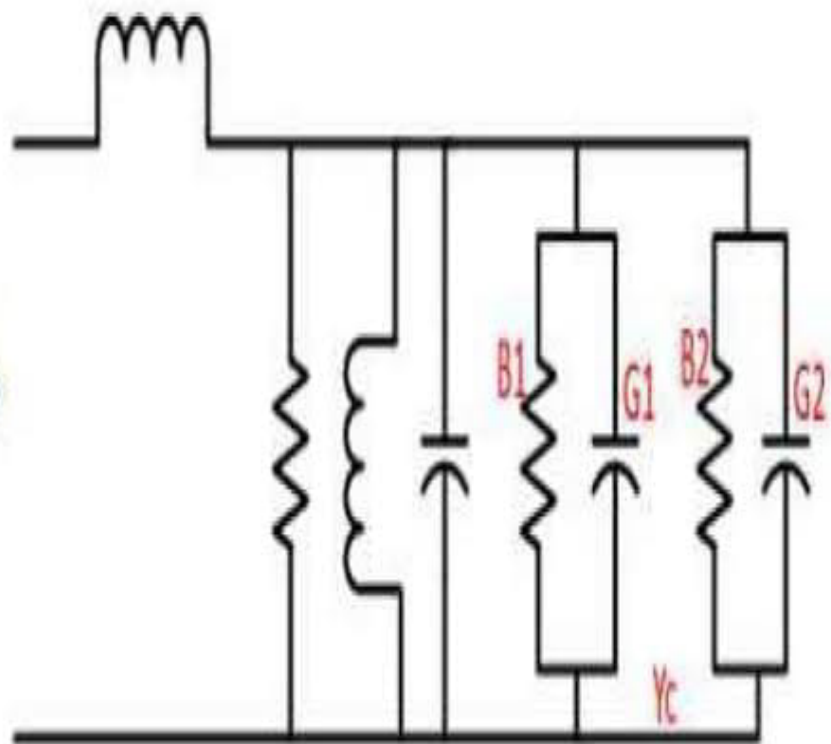
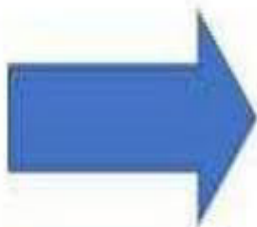
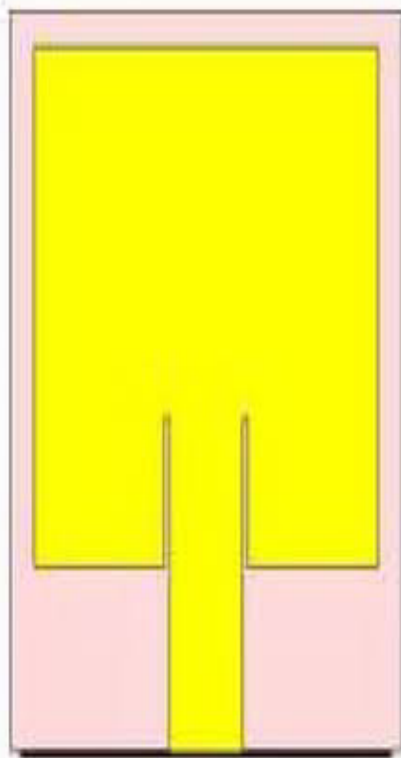
Width of patch depends on dielectric constant and frequency

$$2. \quad \epsilon_e = \frac{\epsilon + 1}{2} + \frac{\epsilon - 1}{2} \left(1 + 12 \left(\frac{h}{W}\right)\right)^{-\frac{1}{2}}$$

Effective dielectric constant accounts for fringing effects

$$3. \quad L = L_{eff} - 2\delta L$$

Actual length is reduced due to edge effects



Methodology

1. Calculate patch dimensions using design dimensions and equations
2. Create a model in ANSYS/antenna magus
3. Assign substrate material(FR4)
4. Apply boundary conditions
5. Excite antenna using feed
6. Simulate and extract parameters

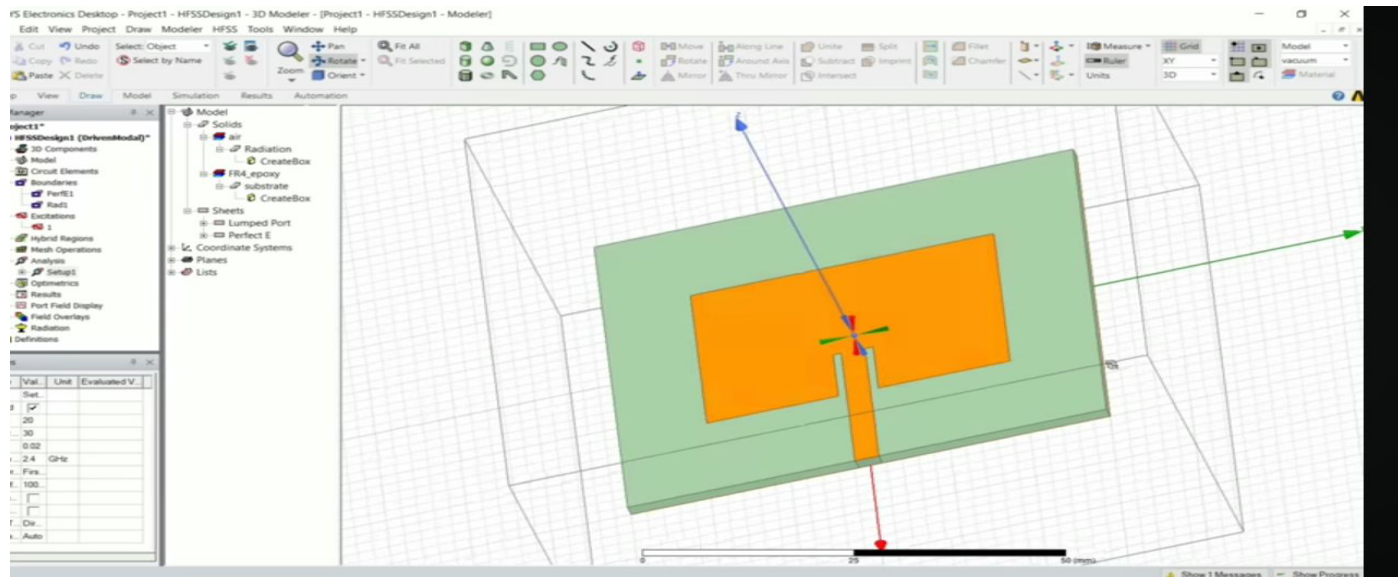
Parameters Analyzed

- Return Loss(S11): Indicates reflected power
Lower value=better matching
Ideal: $s_{11} < -10\text{dB}$
- Bandwidth: Frequency range where antenna works properly
Higher bandwidth is preferred
- Gain: Ability of antenna to radiate power
Measured in dB
Higher gain->better performance
- Directivity: Directional radiation capability
Indicates focus of radiated energy

Key Results

- Return Loss(S_{11}) analysis
- S_{11} indicates reflected power from antenna
- Good antenna condition: $S_{11} < -10\text{dB}$
- Simulated antenna resonates near operating frequency
- Lower S_{11} value indicates better impedance matching
- Minimum S_{11} obtained at resonant frequency
- Efficiency radiation achieved
- Ideal condition: $VSWR = 1$
- Indicates radiated power capability
- Higher gain improves communication range

Outputs



Conclusion

- Microstrip patch antenna performance depends on substrate
- Lower dielectric constant -> better bandwidth
- Higher dielectric constant -> better gain
- Proper design improves antenna efficiency
- Uniform current flow improves antenna performance
- Antenna suitable for wireless communication applications
- Gain observed around 4-5 dB

References

- 1.K. Vidhya et al., "Design and Analysis of Rectangular Microstrip Patch Antenna," *International Journal of Psychosocial Rehabilitation*, 2020.
- 2.Shreyashree Verma et al., "Design and Performance Analysis of a Rectangular Microstrip Patch Antenna for Wireless Communication at 2.4 GHz," *IJRITCC Journal*, 2023.
- 3.Sarika Khurana et al., "Design and Simulation of Rectangular Microstrip Patch Antenna for Wireless Communication," *IJSRM Journal*, 2017.
- 4.Sattar Othman Hasan et al., "Design and Performance Analysis of Rectangular Microstrip Patch Antennas Using Different Feeding Techniques for 5G Applications," *IJECE Systems*, 2023.
- 5.C. A. Balanis, *Antenna Theory: Analysis and Design*, Wiley Publication.

Thank you.

Any questions ?